

Ph134, Lecture 8

Properties of Fresnel relations for refl/trans at an interface (from last time)

Coherence theory: relates properties of the light in frequency space to properties of the light as a function of time

- **Relationship between light source's $E(t)$ and output intensity of interferometer**

- **Example: monochromatic input**

Fresnel relations for p (TM) polarization:

$$H_{0i}^{(+)} e^{ik_i \sin \theta_i y} e^{-i\omega_i t} - H_{0r}^{(+)} e^{ik_r \sin \theta_r y} e^{-i\omega_r t} = H_{0t}^{(+)} e^{ik_t \sin \theta_t y} e^{-i\omega_t t}$$

$$E_{0i}^{(+)} \cos \theta_i e^{ik_i \sin \theta_i y} e^{-i\omega_i t} + E_{0r}^{(+)} \cos \theta_r e^{ik_r \sin \theta_r y} e^{-i\omega_r t} = E_{0t}^{(+)} \cos \theta_t e^{ik_t \sin \theta_t y} e^{-i\omega_t t}$$

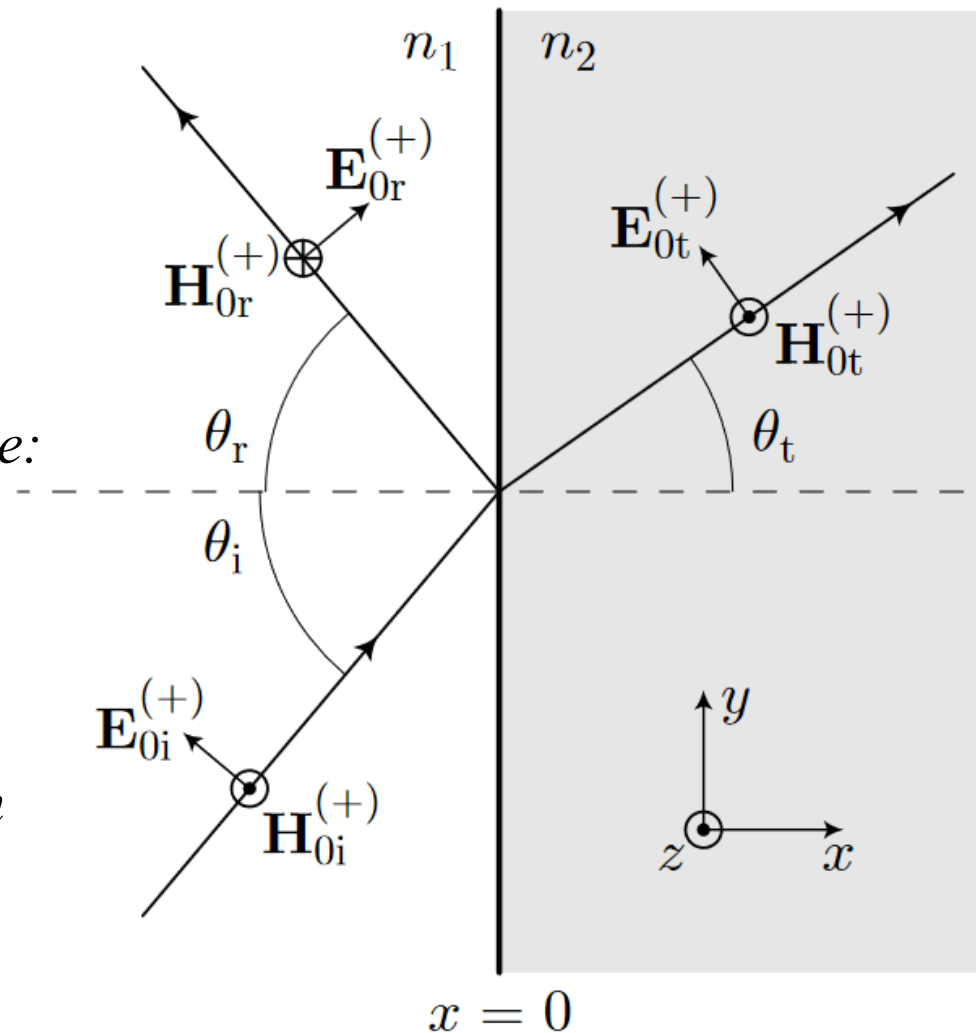
$$r \equiv \frac{E_{or}^{(+)}}{E_{oi}^{(+)}} = \frac{n_1 \cos \theta_t - n_2 \cos \theta_i}{n_1 \cos \theta_t + n_2 \cos \theta_i}$$

Reflected frac. of intensity incident on interface:

$$R = |r|^2 = \left| \frac{n_1 \cos \theta_t - n_2 \cos \theta_i}{n_1 \cos \theta_t + n_2 \cos \theta_i} \right|^2$$

Brewster's angle: no reflection of p pol. when

$$\theta_i + \theta_t = \frac{\pi}{2} \quad \tan \theta_i = \frac{n_2}{n_1}$$



Fresnel relations for s (TE) polarization:

Done in detail in Steck 9.1.

Some things the same:

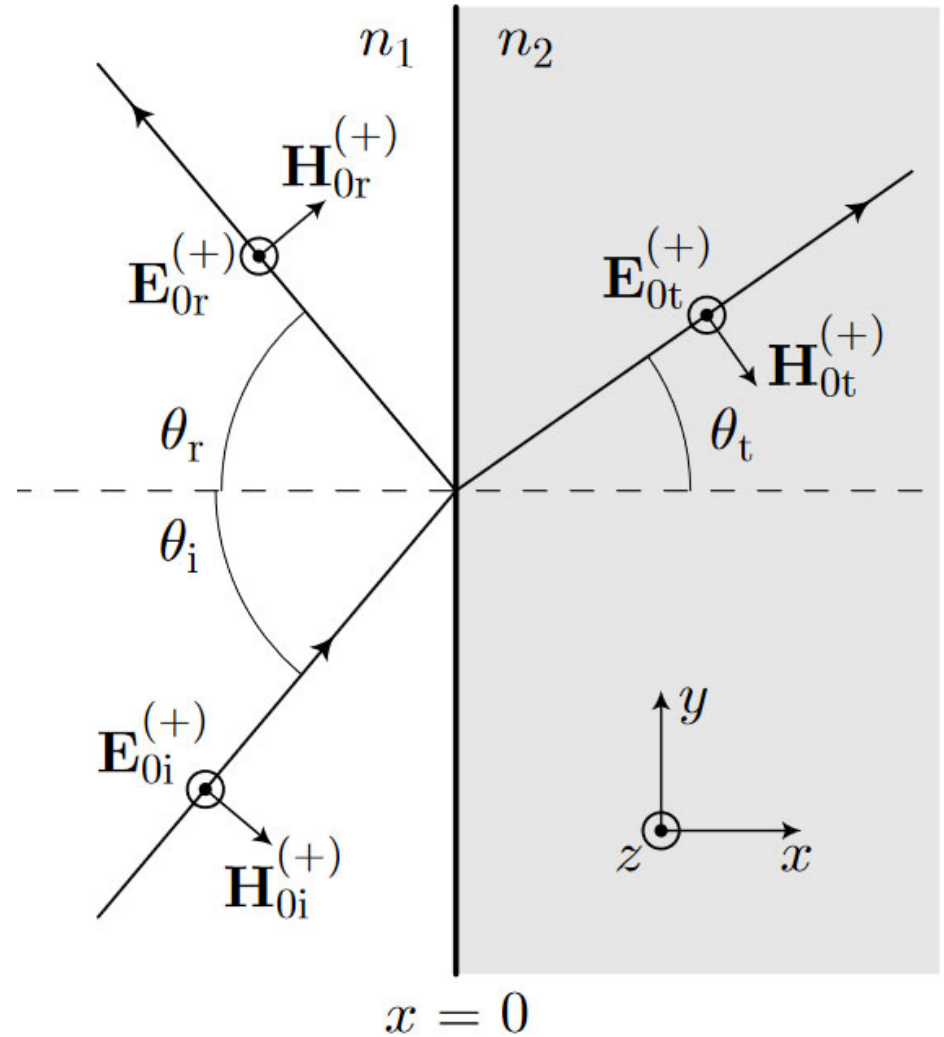
$$\omega_i = \omega_r = \omega_t$$

$$\theta_r = \theta_i$$

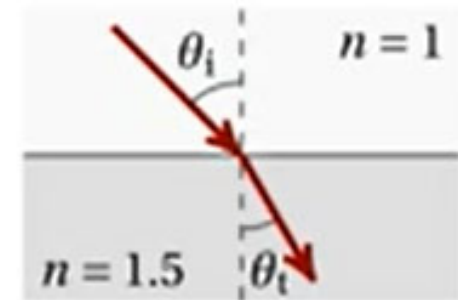
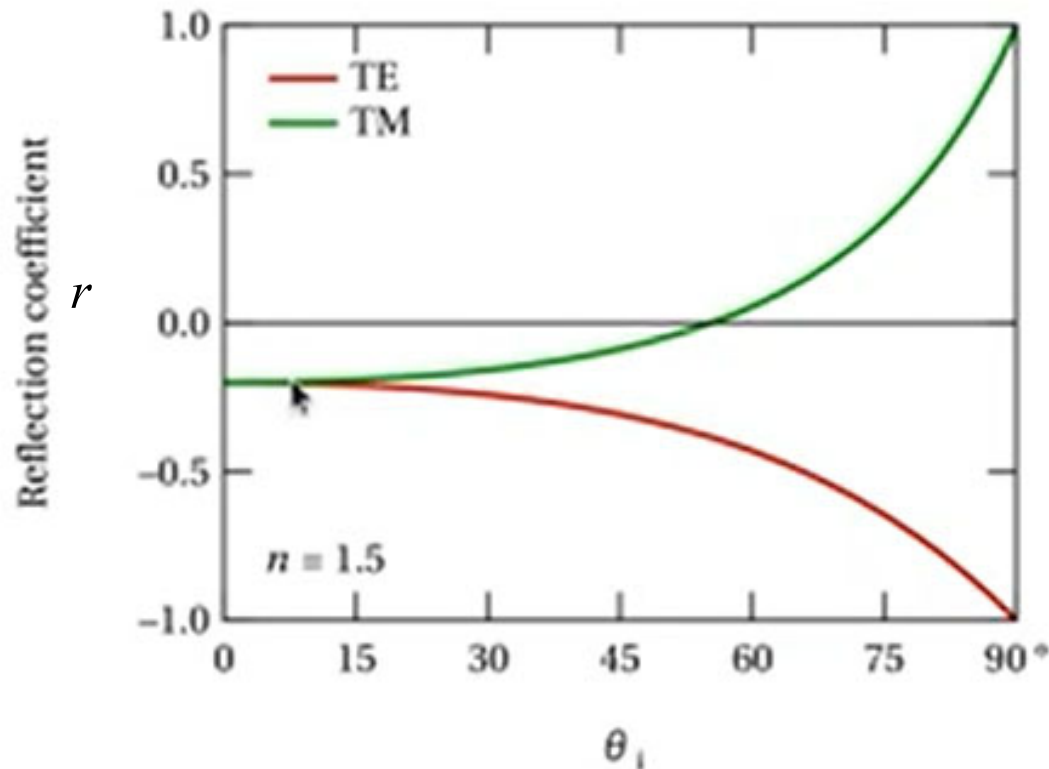
$$n_1 \sin \theta_i = n_2 \sin \theta_t$$

*Reflection and transmission coefficients different.
(Thus reflectivity different.)*

S polarization:



Example reflection coefficients (air to glass):



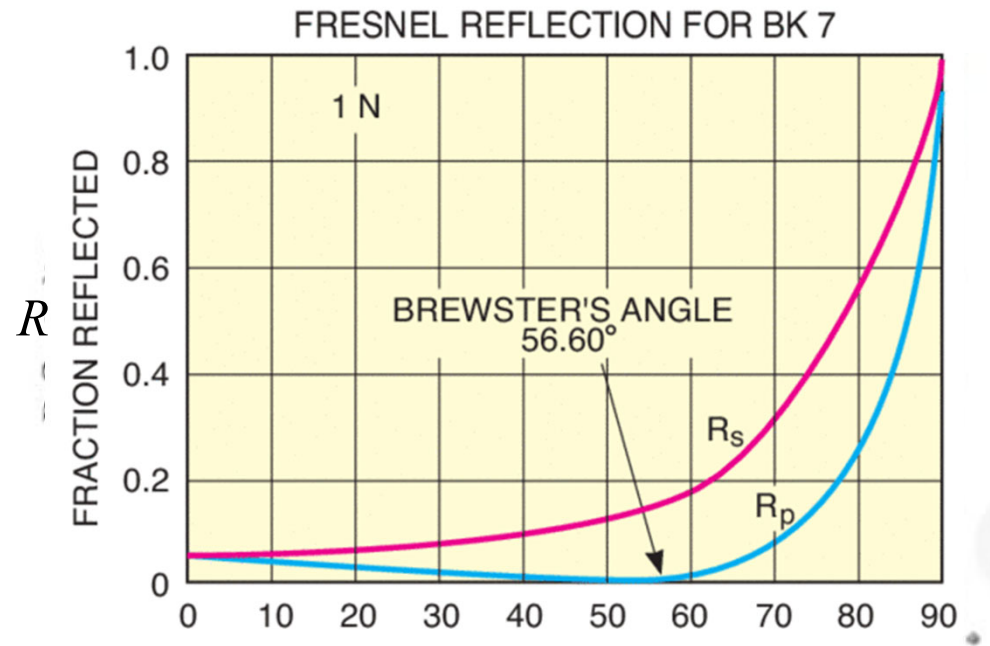
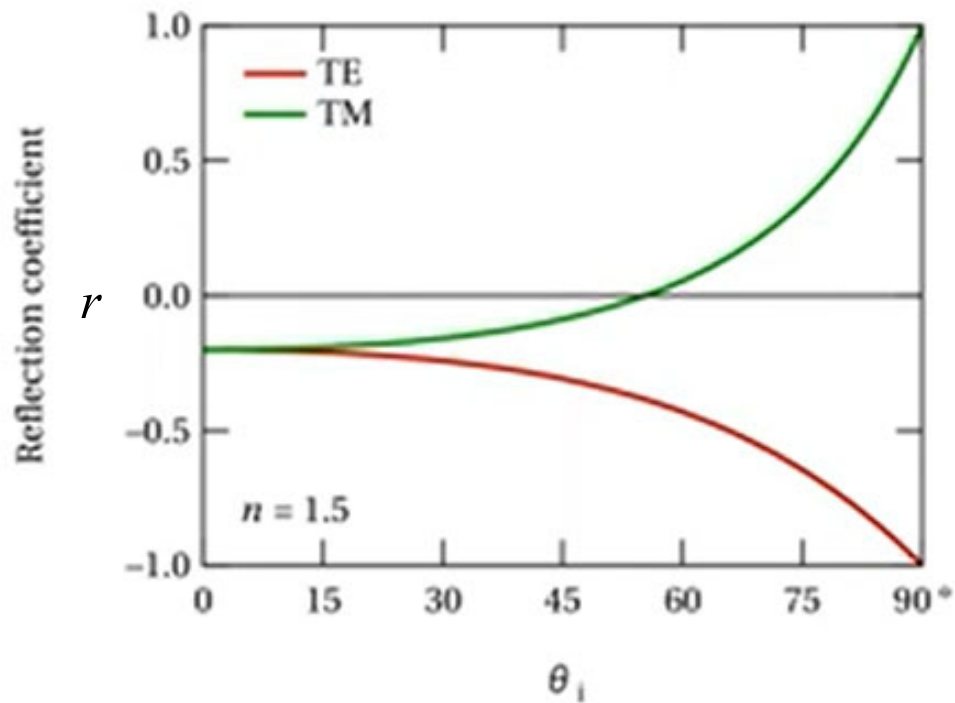
Reflected and transmitted fractions of intensity incident on interface:

$$R = |r|^2$$

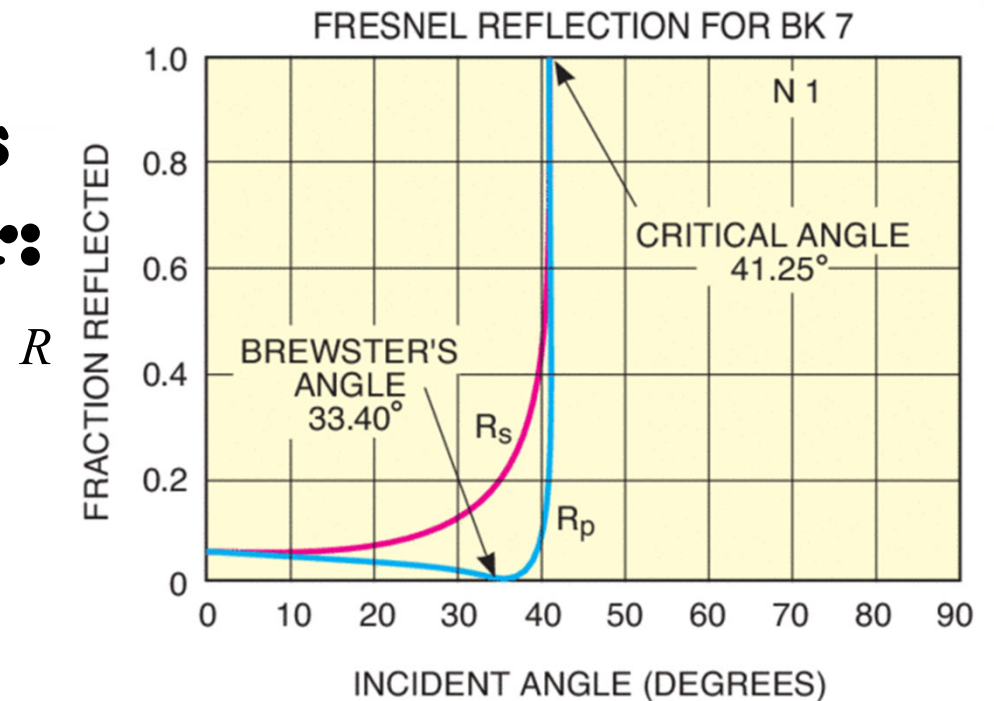
$$T = 1 - R$$

Reflectivity

Air to glass:

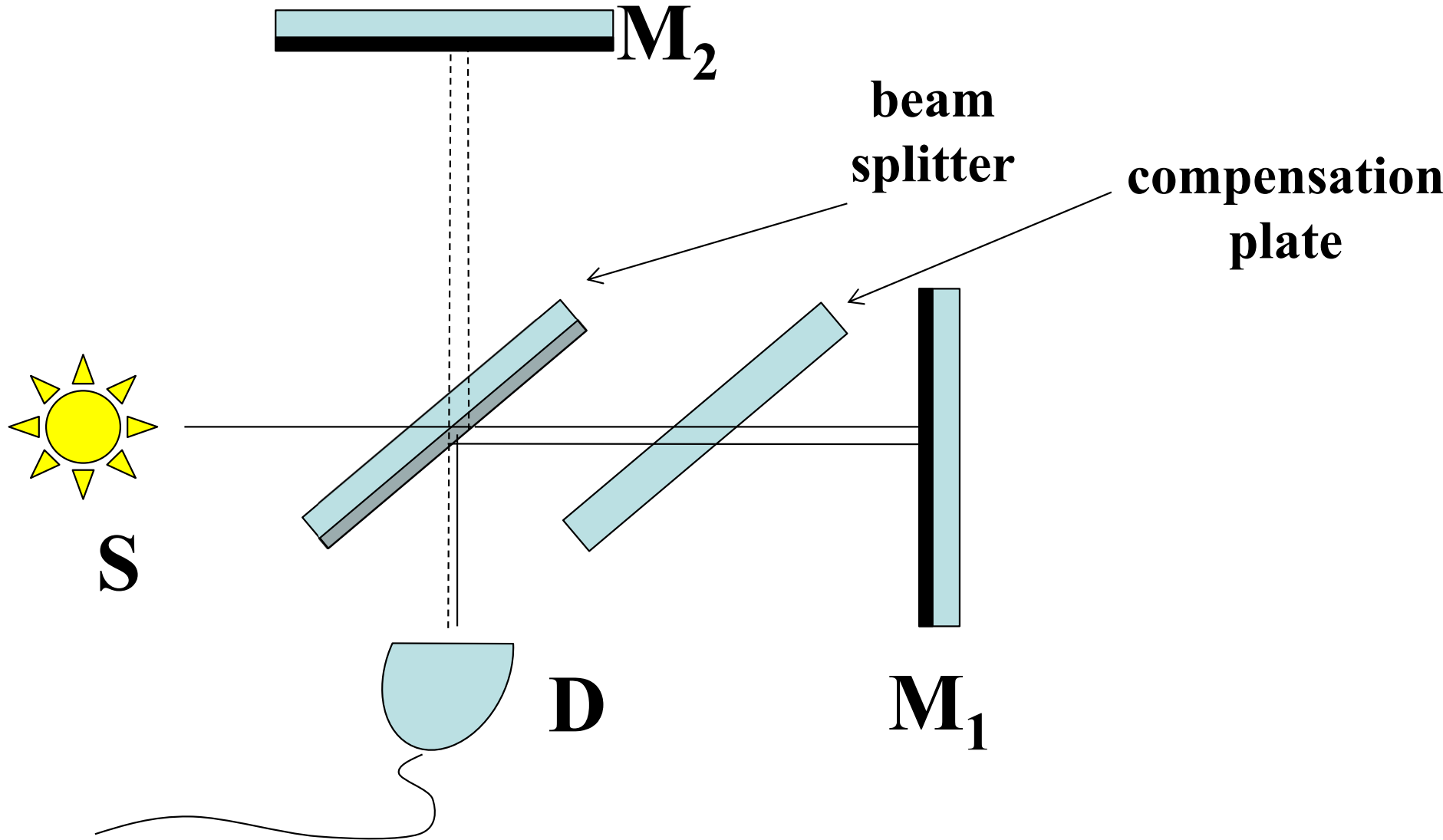


Glass to air:

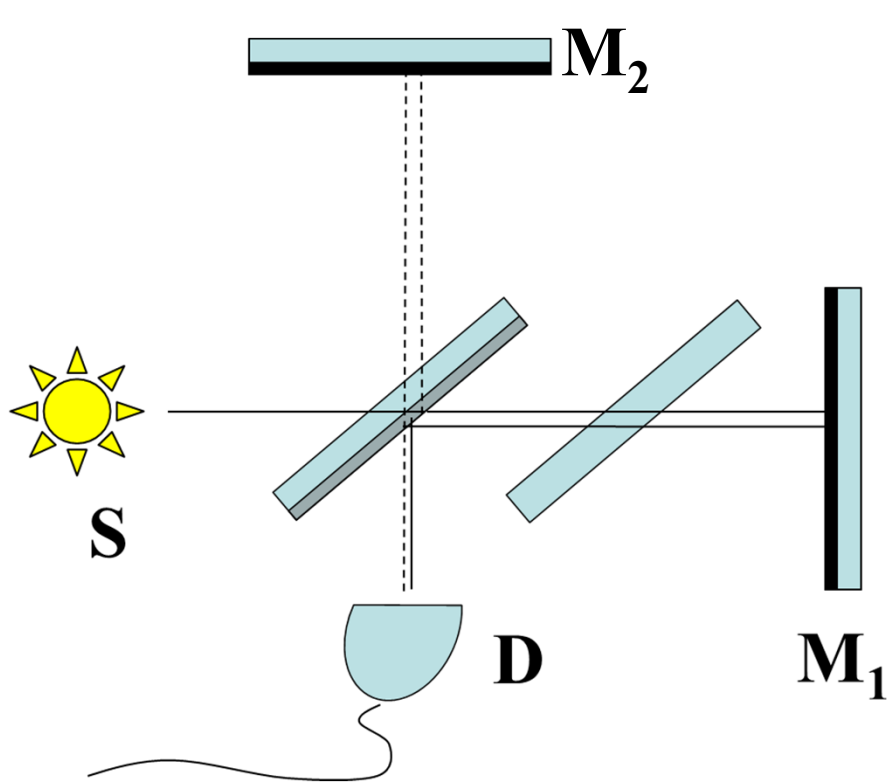


Coherence theory:
relates properties of the light in
frequency space to properties of
the light as a function of time

Consider a Michelson interferometer:



Output vs. input intensities:



$E(t)$ = scalar electric field of light leaving S at time t (consider one polarization for simplicity)

t_1 = time from S to D via path 1

t_2 = time from S to D via path 2

$\tau = t_2 - t_1$

I_{IN} = intensity at S

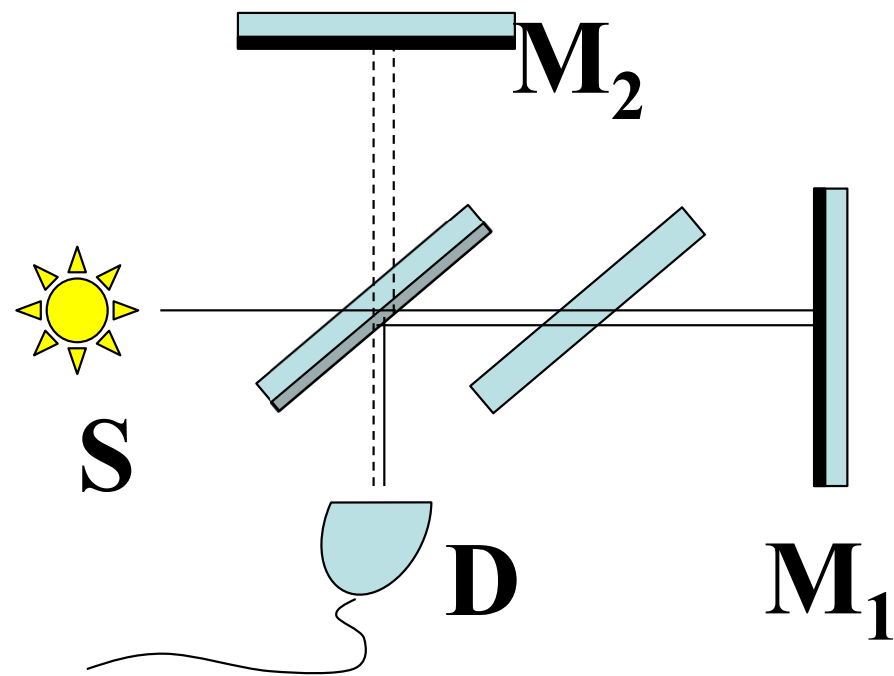
I_{OUT} = intensity at D

$$I_{OUT} = \frac{1}{\mu_0 c} \left[\langle E^{(+)}(t) E^{(-)}(t) \rangle - \text{Re} \langle E^{(+)}(t + \tau) E^{(-)}(t) \rangle \right]$$

$$I_{OUT} = \frac{1}{2} I_{IN} - \frac{1}{\mu_0 c} \text{Re} \langle E^{(+)}(t + \tau) E^{(-)}(t) \rangle = \frac{1}{2} I_{IN} [1 - \text{Re } g^{(1)}(\tau)]$$

where $g^{(1)}(\tau) \equiv \frac{\langle E^{(+)}(t + \tau) E^{(-)}(t) \rangle}{\langle E^{(+)}(t) E^{(-)}(t) \rangle}$

Monochromatic source:



Complete interference fringe

when τ changes by $\frac{2\pi}{\omega}$.

$$\tau = \frac{\text{path length 2} - \text{path length 1}}{c}$$

Complete interference fringe

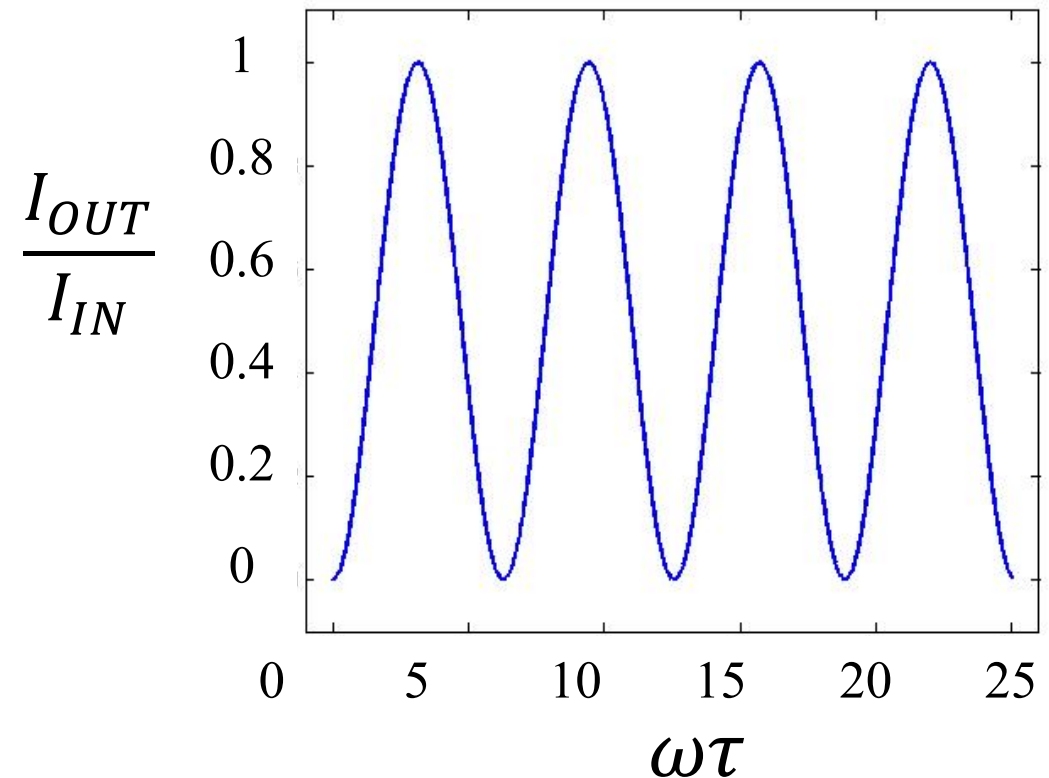
when path length difference changes by λ .

$$E(t) = E_0^{(+)} e^{-i\omega t} + E_0^{(-)} e^{i\omega t}$$

$$I_{OUT} = \frac{1}{2} I_{IN} [1 - \text{Re } g^{(1)}(\tau)]$$

$$\text{where } g^{(1)}(\tau) \equiv \frac{\langle E^{(+)}(t+\tau) E^{(-)}(t) \rangle}{\langle E^{(+)}(t) E^{(-)}(t) \rangle}$$

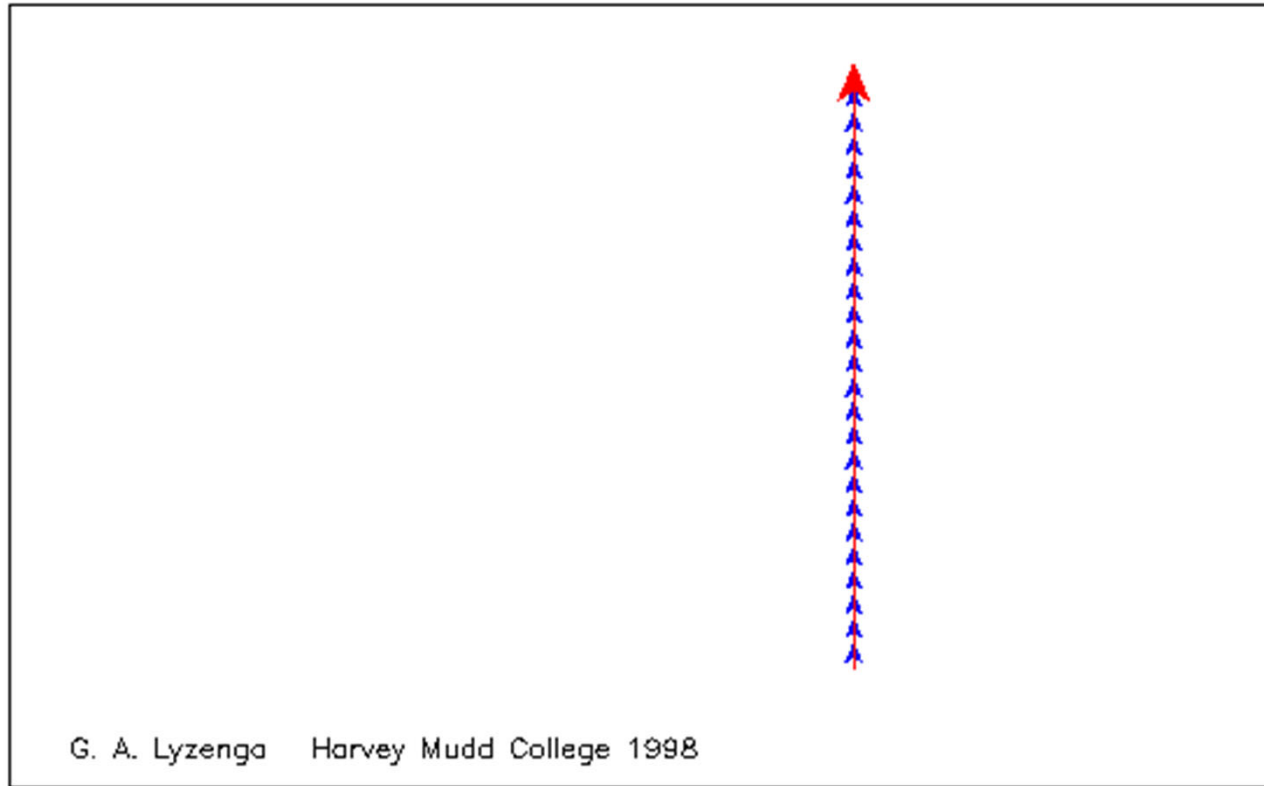
$$I_{OUT} = \frac{I_{IN}}{2} (1 - \cos \omega \tau)$$



Sharper interference maxima?

Interfere N paths with equal delay τ between successive paths:

$$N=25$$



You don't get $N > 2$ in a Michelson interferometer, but this is the physics behind the excellent resolution of the Fabry-Perot interferometer or the grating spectrometer.